

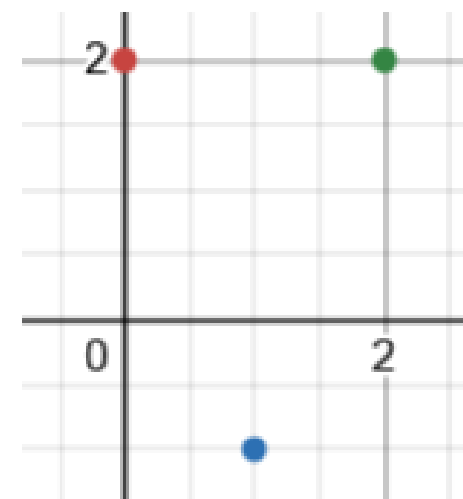
7 – Theoretical transformations

0	2
1	-1
2	2

Row or observation or data *point*

0	2
1	-1
2	2

Row or observation or data *point*

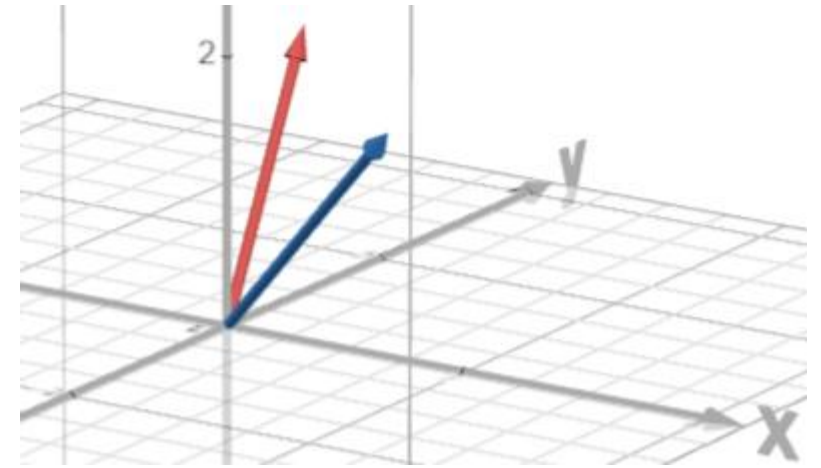


Column or variable or data *vector*

0	2
1	-1
2	2

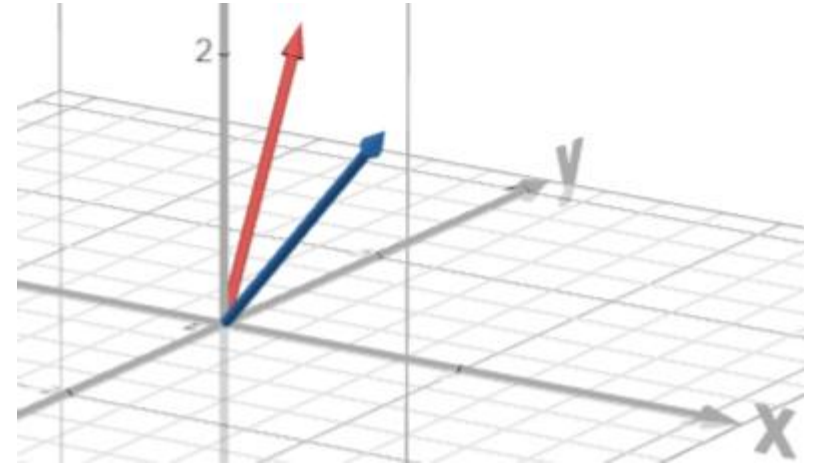
Column or variable or data *vector*

0	2
1	-1
2	2



Column or variable or data *vector*

0	2
1	-1
2	2



Vectors have **length** given by Euclidean distance (L2-norm, Pythagorean theorem):

$$\|(0,1,2)\|_2 = \sqrt{0^2 + 1^2 + 2^2}$$

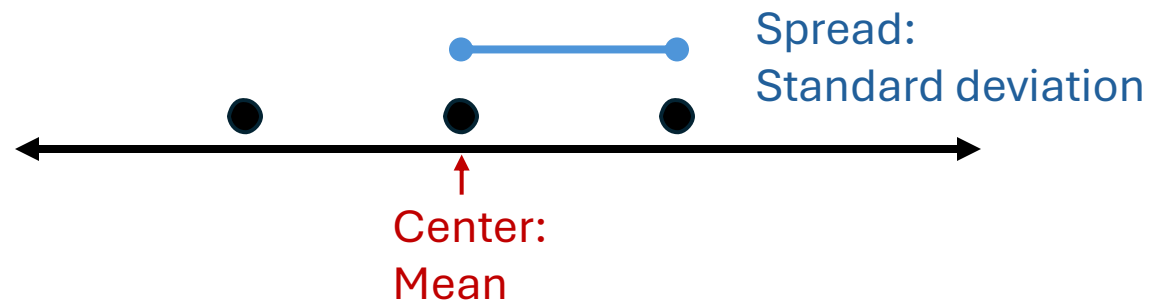
Univariate numeric data:

0
1
2

Univariate numeric data:

0
1
2

Visualize data points as dotplot, histogram, boxplot, etc.



Univariate numeric data:

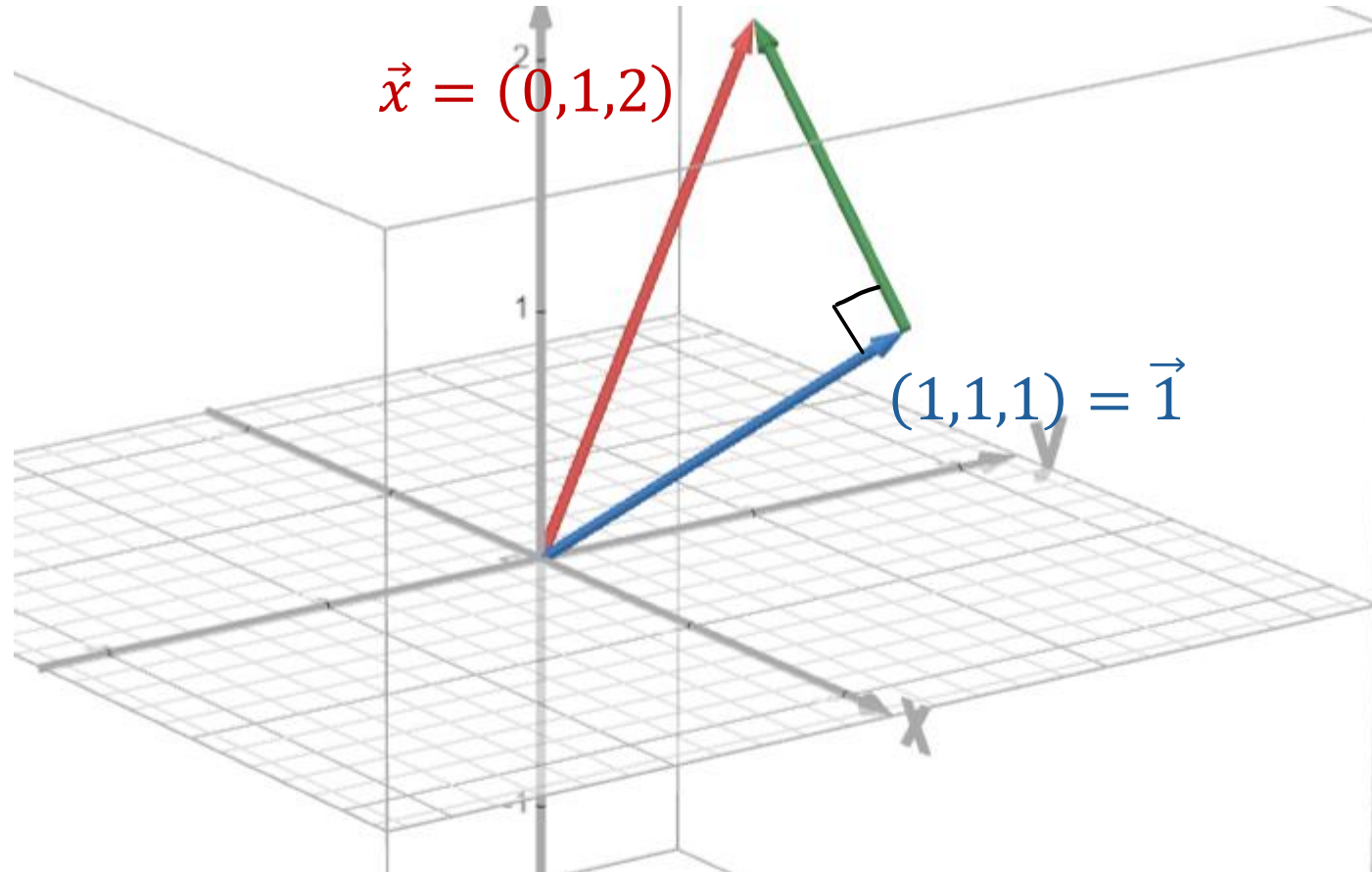
View data vector relative to the homogeneous all-ones vector:

0
1
2

Univariate numeric data:

View data vector relative to the homogeneous all-ones vector:

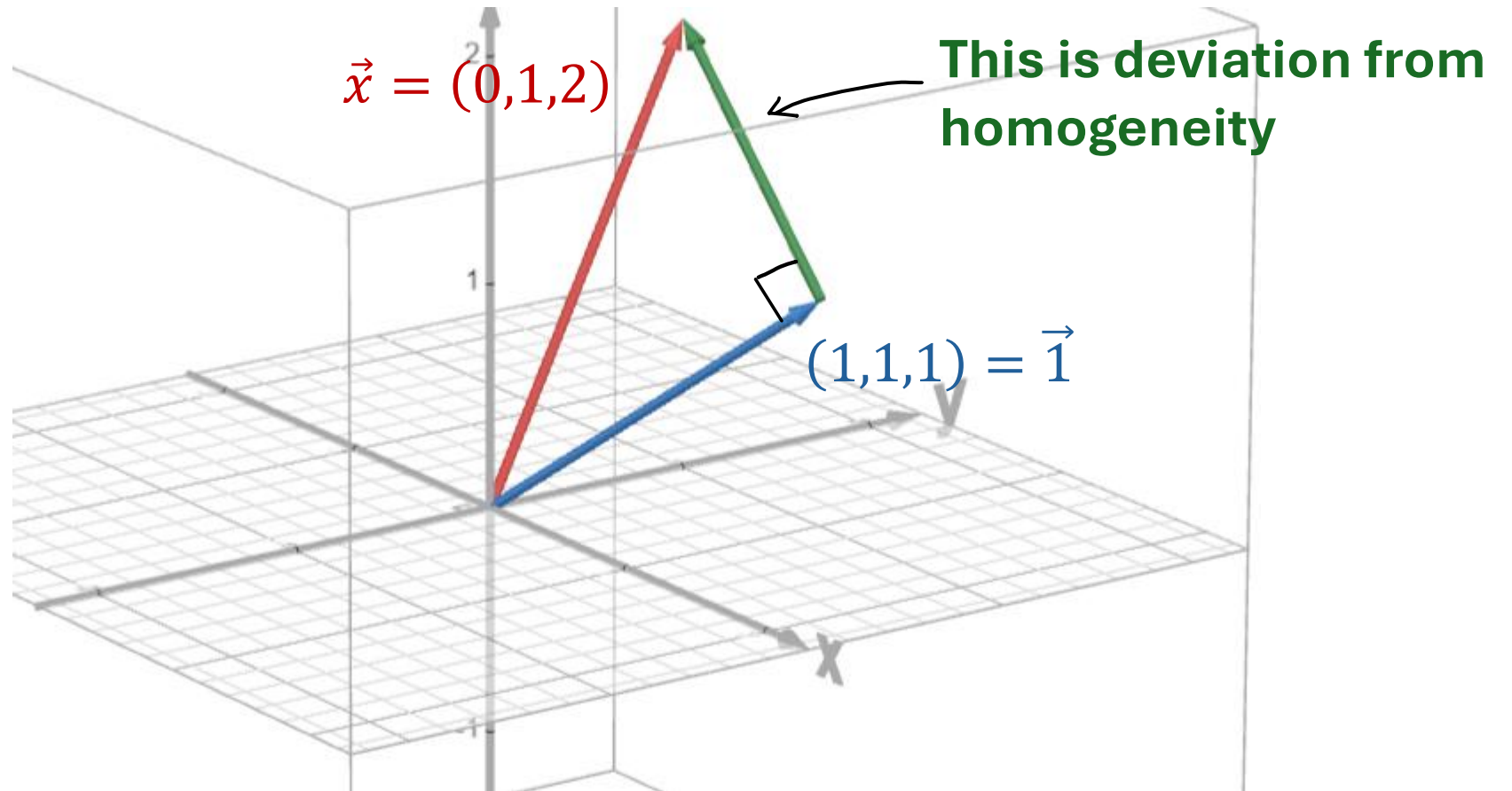
0
1
2



Univariate numeric data:

View data vector relative to the homogeneous all-ones vector:

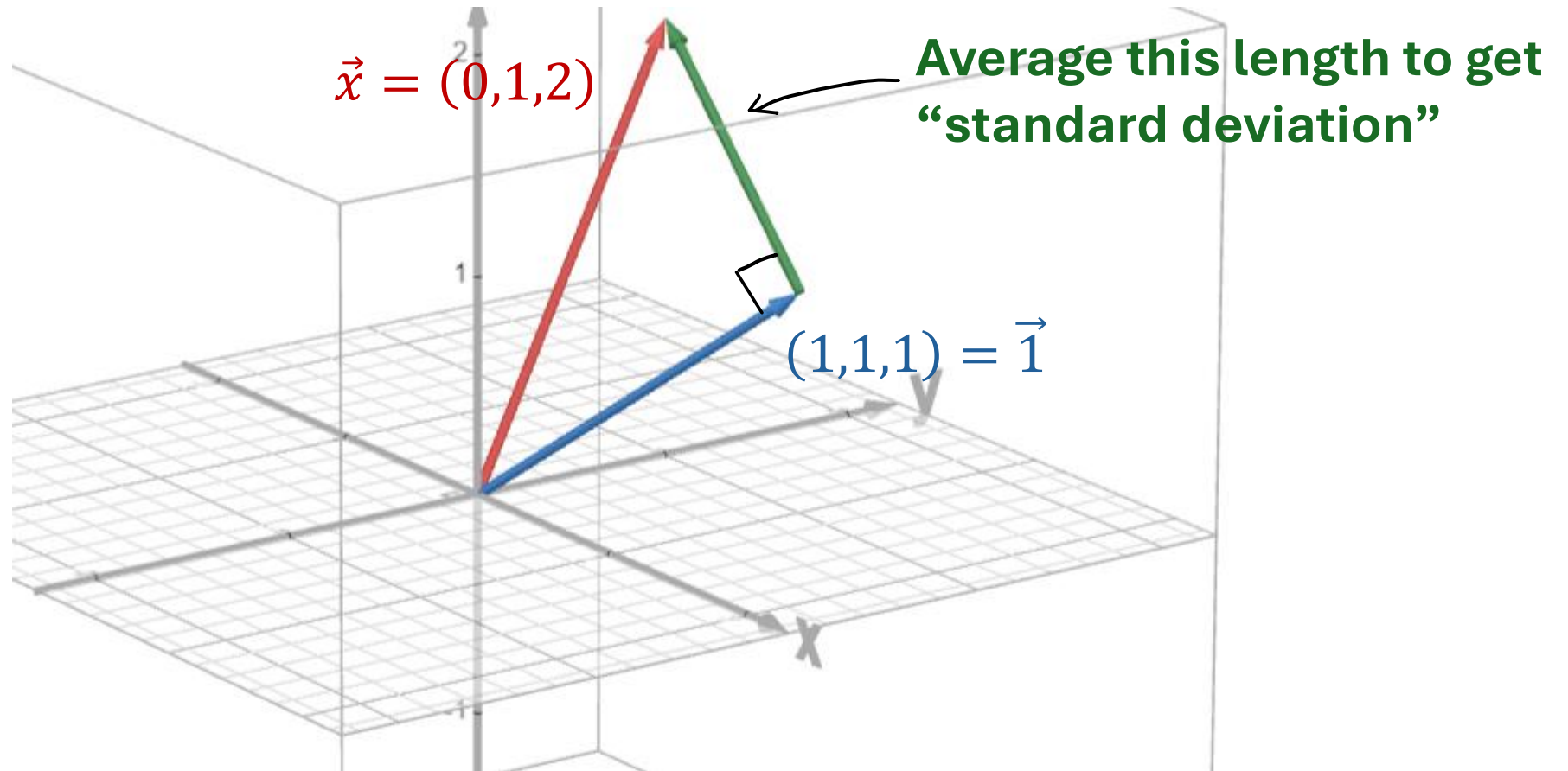
0
1
2



Univariate numeric data:

View data vector relative to the homogeneous all-ones vector:

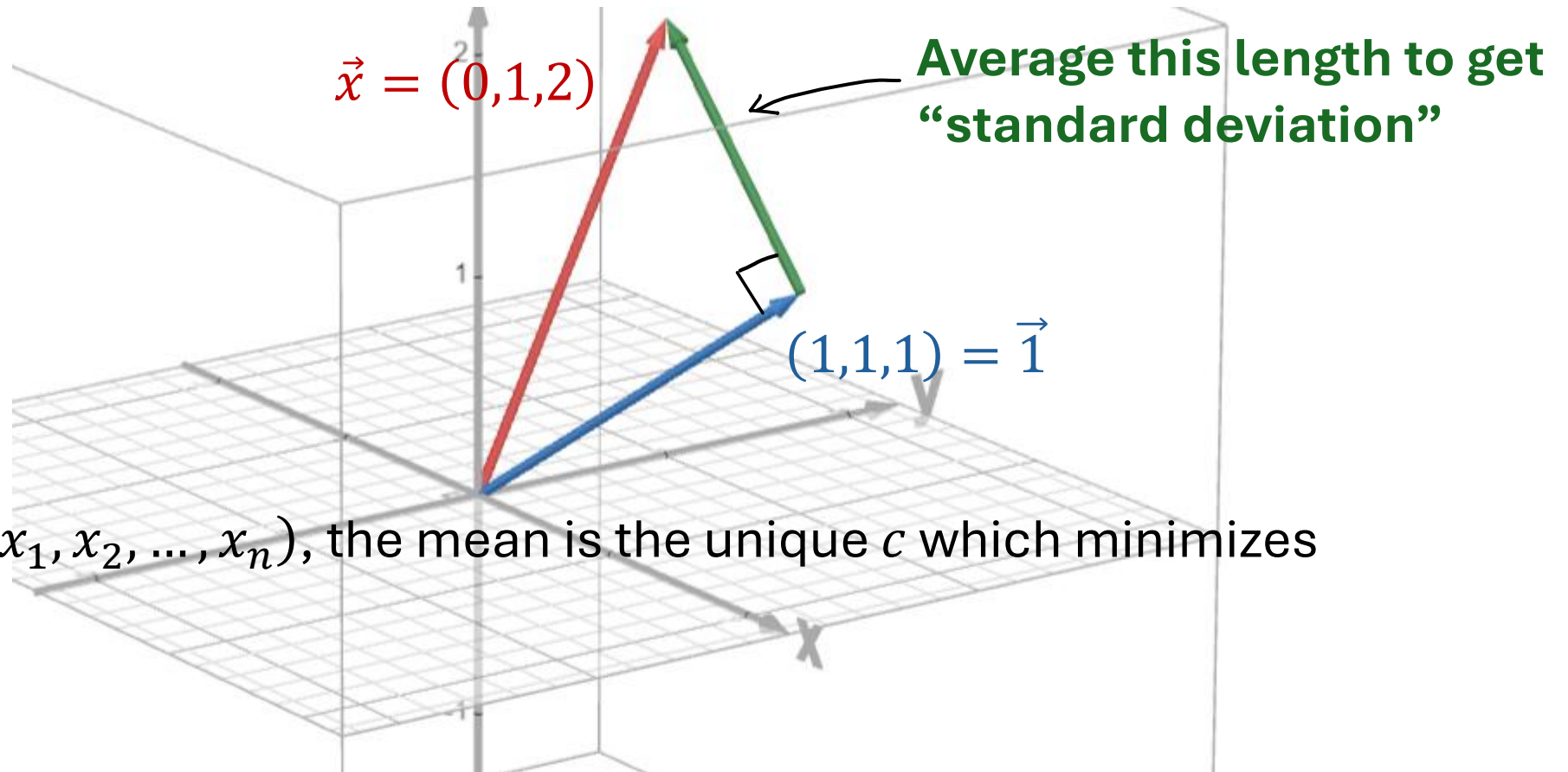
0
1
2



Univariate numeric data:

View data vector relative to the homogeneous all-ones vector:

0
1
2

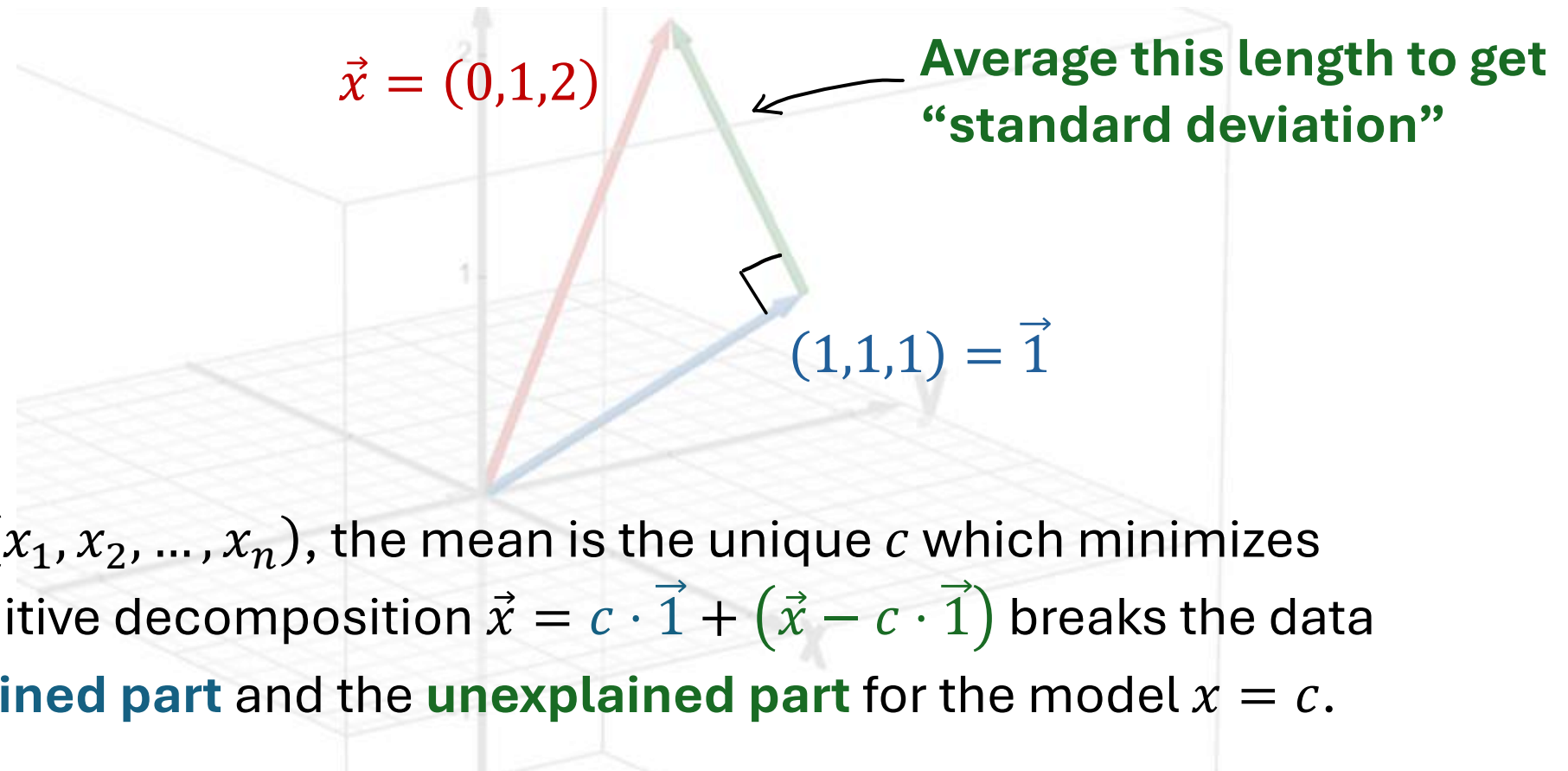


For data vector $\vec{x} = (x_1, x_2, \dots, x_n)$, the mean is the unique c which minimizes $\|\vec{x} - c \cdot \vec{1}\|_2$.

Univariate numeric data:

View data vector relative to the homogeneous all-ones vector:

0
1
2



For data vector $\vec{x} = (x_1, x_2, \dots, x_n)$, the mean is the unique c which minimizes $\|\vec{x} - c \cdot \vec{1}\|_2$. The additive decomposition $\vec{x} = c \cdot \vec{1} + (\vec{x} - c \cdot \vec{1})$ breaks the data vector into the **explained part** and the **unexplained part** for the model $x = c$.

Univariate numeric data:

This decomposition breaks our vector space of data vectors into two parts:

$$\mathbb{R}^n = \mathbb{R}^1 \oplus \mathbb{R}^{n-1}$$
$$\vec{x} = \bar{x} \cdot \vec{1} + (\vec{x} - \bar{x} \cdot \vec{1})$$

Univariate numeric data:

This decomposition breaks our vector space of data vectors into two parts:

$$\mathbb{R}^n = \mathbb{R}^1 \oplus \mathbb{R}^{n-1}$$
$$\vec{x} = \bar{x} \cdot \vec{1} + (\vec{x} - \bar{x} \cdot \vec{1})$$

The **uncorrected sample variance** is $\frac{1}{n} \|\vec{x} - \bar{x} \cdot \vec{1}\|_2^2$.

Its square root is the **uncorrected sample standard deviation**.

Univariate numeric data:

This decomposition breaks our vector space of data vectors into two parts:

$$\mathbb{R}^n = \mathbb{R}^1 \oplus \mathbb{R}^{n-1}$$
$$\vec{x} = \bar{x} \cdot \vec{1} + (\vec{x} - \bar{x} \cdot \vec{1})$$

The **uncorrected sample variance** is $\frac{1}{n} \|\vec{x} - \bar{x} \cdot \vec{1}\|_2^2$.

Its square root is the **uncorrected sample standard deviation**.

The **(corrected) sample variance** is $\frac{1}{n-1} \|\vec{x} - \bar{x} \cdot \vec{1}\|_2^2$.

Its square root is the **(corrected) sample standard deviation**.

Bivariate numeric data:

0	2
1	-1
2	2

View homogeneous data vector as “explained”.

We first **mean-center** our data vectors to get unexplained part.

We then **standardize** the data vectors to compare them.

Bivariate numeric data:

data
frame

x	y
0	2
1	-1
2	2

View homogeneous data vector as “explained”.

We first **mean-center** our data vectors to get unexplained part.

We then **standardize** the data vectors to compare them.

Code:

```
df$x <- df$x - mean(df$x)      # mean centering  
df$x <- df$x/sd(df$x)         # standardize
```

Bivariate numeric data:

data frame	x	y
	0	2
	1	-1
	2	2

View homogeneous data vector as “explained”.

We first **mean-center** our data vectors to get unexplained part.

We then **standardize** the data vectors to compare them.

Code:

```
df$x <- df$x - mean(df$x)      # mean centering  
df$x <- df$x/sd(df$x)         # standardize
```

Bivariate numeric data:

data frame	x	y
	-1	1
	0	-2
	1	1

View homogeneous data vector as “explained”.

We first **mean-center** our data vectors to get unexplained part.

We then **standardize** the data vectors to compare them.

Code:

```
df$x <- df$x - mean(df$x)      # mean centering  
df$x <- df$x/sd(df$x)          # standardize
```

Bivariate numeric data:

$\vec{x}_{\text{std}}$	$\vec{y}_{\text{std}}$
-1	0.58
0	-1.15
1	0.58

View homogeneous data vector as “explained”.

We first **mean-center** our data vectors to get unexplained part.

We then **standardize** the data vectors to compare them.

Code:

```
df$x <- df$x - mean(df$x)      # mean centering  
df$x <- df$x/sd(df$x)         # standardize
```

Bivariate numeric data:

$\vec{x}_{\text{std}}$	$\vec{y}_{\text{std}}$
-1	0.58
0	-1.15
1	0.58

View homogeneous data vector as “explained”.

We first **mean-center** our data vectors to get unexplained part.

We then **standardize** the data vectors to compare them.

Code:

```
df$x <- df$x - mean(df$x)      # mean centering  
df$x <- df$x/sd(df$x)         # standardize
```

0.58 is the **z-score** of the data value 2 in the data vector **y**.

Bivariate numeric data:

$\vec{x}_{\text{std}}$	$\vec{y}_{\text{std}}$
-1	0.58
0	-1.15
1	0.58

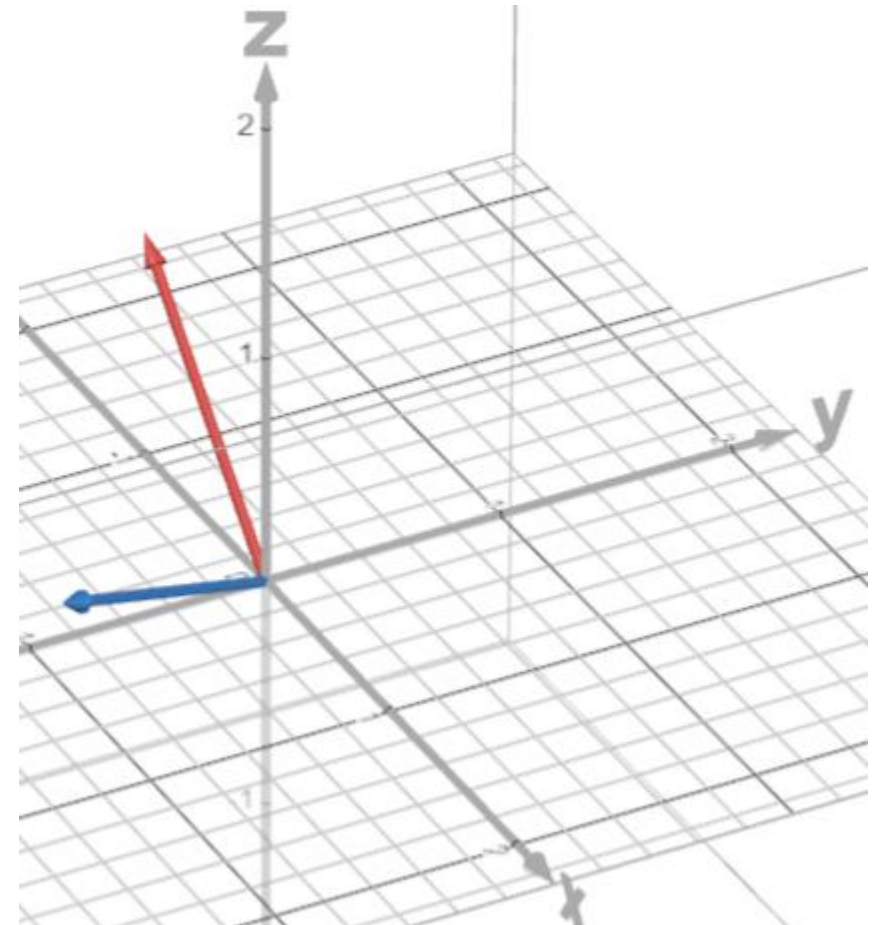
On data points:



Best fit line: horizontal (uncorrelated)

On data vectors:

Perpendicular



Bivariate numeric data:

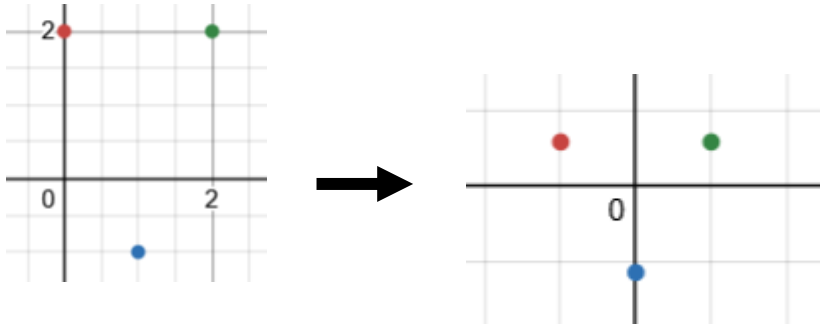
$\vec{x}_{\text{std}}$	$\vec{y}_{\text{std}}$
-1	0.58
0	-1.15
1	0.58

Fact: Ordinary least squares (OLS) regression line is

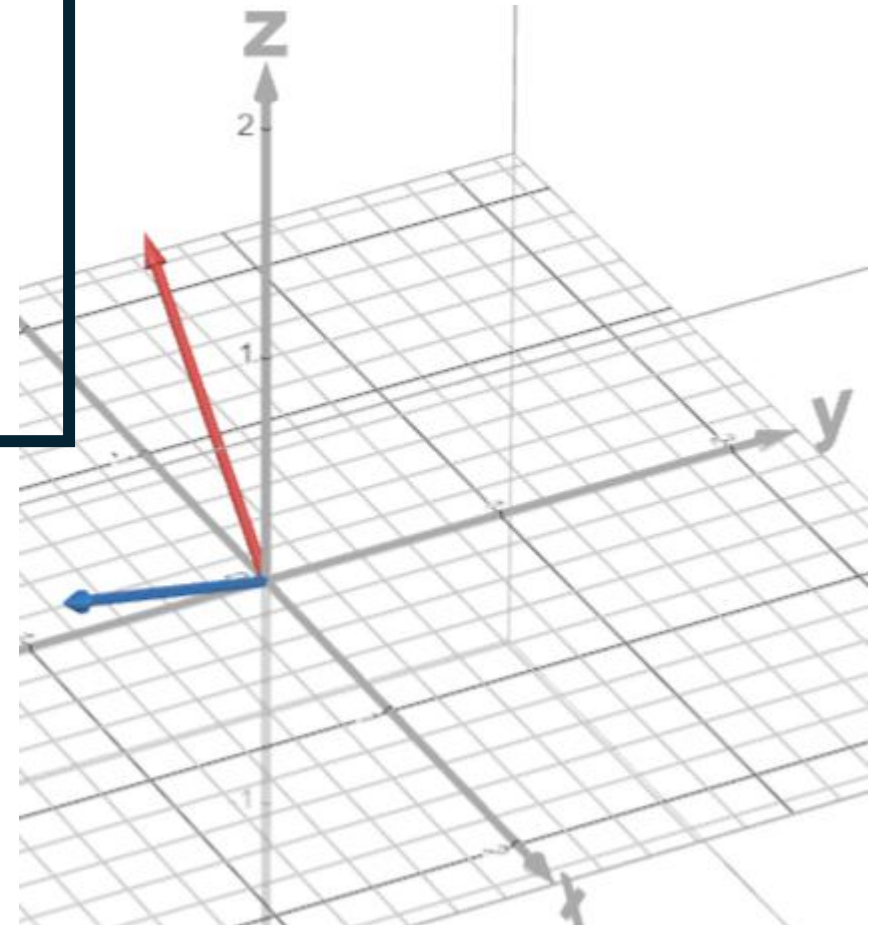
$$y_{\text{std}} = (\cos \theta) \cdot x_{\text{std}}$$

where θ is angle between $\vec{x}_{\text{std}}$ and $\vec{y}_{\text{std}}$.

On data points:



On data vectors:



Bivariate numeric data:

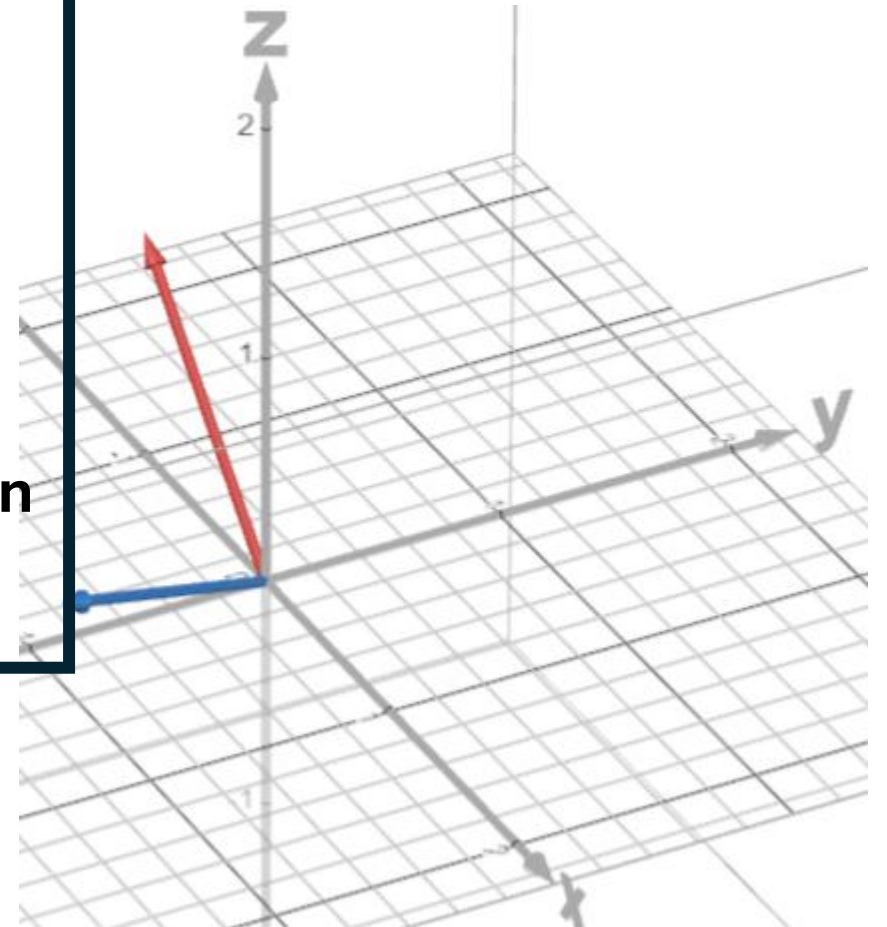
$\vec{x}_{\text{std}}$	$\vec{y}_{\text{std}}$
-1	0.58
0	-1.15
1	0.58

Fact: Ordinary least squares (OLS) regression line is

$$y_{\text{std}} = (\cos \theta) \cdot x_{\text{std}}$$

where θ is angle between $\vec{x}_{\text{std}}$ and $\vec{y}_{\text{std}}$.

$\cos \theta$ is the **(Pearson) correlation** of $\vec{x}$ and $\vec{y}$ and denoted r_{xy} or r .



Collaborative filtering

Movies rated	Mov 1	Mov 2	Mov 3	Movie 4	Movie 5	Movie 6	Movie 7	Movie 8
User u_1	5	1	4	.	.	.	2	1
User u_2	5	.	3	2	5	.	2	1
User u_3	1	4	2	5	2	.	5	4
User u_4	4	2	.	1	5	2	3	3

Collaborative filtering

Movies rated	1	2	3	4	5	6	7	8
User u_1	5	1	4	.	.	.	2	1
User u_2	5	.	3	2	5	.	2	1
User u_3	1	4	2	5	2	.	5	4
User u_4	4	2	.	1	5	2	3	3

User most like u_1 ? Compute user-user correlation, ignoring NA values via the use option:

Collaborative filtering

Movies rated	1	2	3	4	5	6	7	8
User u_1	5	1	4	.	.	.	2	1
User u_2	5	.	3	2	5	.	2	1
User u_3	1	4	2	5	2	.	5	4
User u_4	4	2	.	1	5	2	3	3

User most like u_1 ? Compute user-user correlation, ignoring NA values via the use option:

```
cor(df$u1, df$u2, use = "pairwise.complete.obs")
```

```
cor(df$u1, df$u3, use = "pairwise.complete.obs")
```

```
cor(df$u1, df$u4, use = "pairwise.complete.obs")
```

Collaborative filtering

Movies rated	1	2	3	4	5	6	7	8
User u_1	5	1	4	.	.	.	2	1
User u_2	5	.	3	2	5	.	2	1
User u_3	1	4	2	5	2	.	5	4
User u_4	4	2	.	1	5	2	3	3

User most like u_1 ? Compute user-user correlation, ignoring NA values via the use option:

`cor(df$u1, df$u2, use = "pairwise.complete.obs") = 0.96 > 0 (like-minded)`

`cor(df$u1, df$u3, use = "pairwise.complete.obs") = -0.89`

`cor(df$u1, df$u4, use = "pairwise.complete.obs") = 0.86 > 0 (like-minded)`

Collaborative filtering

Movies rated	1	2	3	4	5	6	7	8
User u_1	5	1	4	.	.	.	2	1
User u_2	5	.	3	2	5	.	2	1
User u_3	1	4	2	5	2	.	5	4
User u_4	4	2	.	1	5	2	3	3

User most like u_1 ? Compute user-user correlation, ignoring NA values via the use option:

`cor(df$u1, df$u2, use = "pairwise.complete.obs") = 0.96 > 0 (like-minded)`

`cor(df$u1, df$u3, use = "pairwise.complete.obs") = -0.89`

`cor(df$u1, df$u4, use = "pairwise.complete.obs") = 0.86 > 0 (like-minded)`

Infer User 1's preference for movie 4 from like-minded users:

$$u_{1,4} - \text{mean}(u_1) = \frac{r_{12}}{r_{12} + r_{14}} [u_{24} - \text{mean}(u_2)] + \frac{r_{14}}{r_{12} + r_{14}} [u_{44} - \text{mean}(u_4)]$$

Collaborative filtering

Movies rated	1	2	3	4	5	6	7	8
User u_1	5	1	4	.	.	.	2	1
User u_2	5	.	3	2	5	.	2	1
User u_3	1	4	2	5	2	.	5	4
User u_4	4	2	.	1	5	2	3	3

User most like u_1 ? Compute user-user correlation, ignoring NA values via the use option:

`cor(df$u1, df$u2, use = "pairwise.complete.obs") = 0.96 > 0 (like-minded)`

`cor(df$u1, df$u3, use = "pairwise.complete.obs") = -0.89`

`cor(df$u1, df$u4, use = "pairwise.complete.obs") = 0.86 > 0 (like-minded)`

Infer User 1's preference for movie 4 from like-minded users:

$$u_{1,4} - 2.6 = \frac{0.96}{0.96 + 0.86} [2 - 3] + \frac{0.86}{0.96 + 0.86} [1 - 2.9] = -0.5 \Rightarrow u_{1,4} = 2.1$$

Imputation: listwise deletion

Throwing out data with NA's....

But this wastes useful data or introduces bias.

Country	Income	Age
US	14	29
China	18	36
US	NA	53
US	NA	32
China	NA	14
US	20	47
China	20	56
US	14	25

Listwise deletion

```
df_clean <- na.omit(df)
```

Country	Income	Age
US	14	29
China	18	36
■	■	■
■	■	■
■	■	■
US	20	47
China	20	56
US	14	25

Imputation: mean imputation

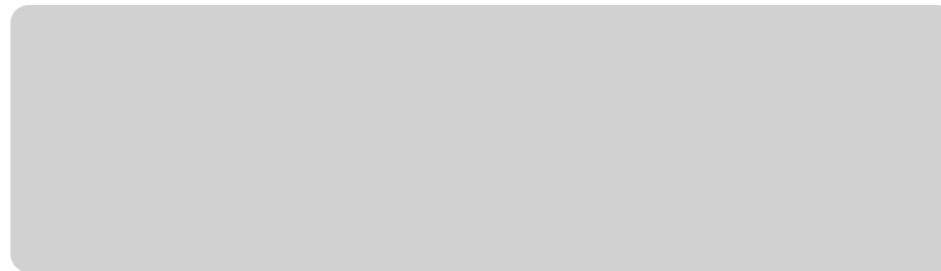
Throwing out data with NA's waste useful data or introduces bias.

Substitute or “impute” missing data with plausible values.

Mean imputation substitutes missing NA's in a column with the column mean.

Country	Income	Age
US	14	29
China	18	36
US	NA	53
US	NA	32
China	NA	14
US	20	47
China	20	56
US	14	25

Mean imputation



Country	Income	Age
US	14	29
China	18	36
US	17.2	53
US	17.2	32
China	17.2	14
US	20	47
China	20	56
US	14	25

Imputation: mean imputation

Throwing out data with NA's waste useful data or introduces bias.

Substitute or “impute” missing data with plausible values.

Mean imputation substitutes missing NA's in a column with the column mean.

Country	Income	Age
US	14	29
China	18	36
US	NA	53
US	NA	32
China	NA	14
US	20	47
China	20	56
US	14	25

Mean imputation

```
df$income_imputed <-  
  ifelse(is.na(df$ income),  
        mean(df$ income, na.rm = TRUE),  
        df$income)
```

Country	Income	Age
US	14	29
China	18	36
US	17.2	53
US	17.2	32
China	17.2	14
US	20	47
China	20	56
US	14	25

Imputation: mean imputation

Throwing out data with NA's waste useful data or introduces bias.

Substitute or “impute” missing data with plausible values.

Mean imputation substitutes missing NA's in a column with the column mean.

Country	Income	Age
US	14	29
China	18	36
US	NA	53
US	NA	32
China	NA	14
US	20	47
China	20	56
US	14	25

(Mean imputation by country)

Groupwise mean imputation



Country	Income	Age
US	14	29
China	18	36
US	16	53
US	16	32
China	19	14
US	20	47
China	20	56
US	14	25

Imputation: mean imputation

Throwing out data with NA's waste useful data or introduces bias.

Substitute or “impute” missing data with plausible values.

Mean imputation substitutes missing NA's in a column with the column mean.

Country	Income	Age
US	14	29
China	18	36
US	NA	53
US	NA	32
China	NA	14
US	20	47
China	20	56
US	14	25

(Mean imputation by country)

Groupwise mean imputation

```
df_clean <- df |>
  group_by(country) |>
  mutate(
    income = ifelse(is.na(income),
      mean(income, na.rm = TRUE), income)
  ) |>
  ungroup()
```

Country	Income	Age
US	14	29
China	18	36
US	16	53
US	16	32
China	19	14
US	20	47
China	20	56
US	14	25

Imputation: mean imputation

Throwing out data with NA's waste useful data or introduces bias.

Substitute or “impute” missing data with plausible values.

Mean imputation substitutes missing NA's in a column with the column mean.

Country	Income	Age
US	14	29
China	18	36
US	NA	53
US	NA	32
China	NA	14
US	20	47
China	20	56
US	14	25

(Mean imputation by country)

Groupwise mean imputation

—————→
Drawback to mean imputation:
underestimate standard deviation

Country	Income	Age
US	14	29
China	18	36
US	16	53
US	16	32
China	19	14
US	20	47
China	20	56
US	14	25